

AMBROSE ALLI UNIVERSITY

EKPOMA

DEPARTMENT OF VOCATIONAL AND TECHNICAL EDUCATION



UNDERGRADUATE PROGRAMMES

2018 -2022



**AMBROSE ALLI UNIVERSITY,
EKPOMA EDO STATE, NIGERIA**

**DEPARTMENT OF VOCATIONAL AND TECHNICAL EDUCATION
HANDBOOK**

FOR

**UNDERGRADUATE STUDENTS
2018-2022**

Name:-----

Matric. No.:----- Level -----

Option:-----

Phone No.:-----

Address:-----

FORWARD

Vocational and Technical Education is one of the most viable discipline offered at the university level to produce individuals that will be self-reliant. In the circumstance, there is the need to provide the students, who have made Vocational and Technical Education their choice firsthand information about this career.

This edition of the handbook, written in conformity with the NUC Benchmark Minimum Academic Standard (BMAS) and the changes taking place in Vocational and Technical Education Programmes worldwide, the content deals with the history, philosophy, objectives of each programme in the Department. The number of staff: academic and non-academic are presented; the administrative structure of the Department as it relates to the university is equally included. The bulk of the presentation relates to students admission requirements, courses content, assessment, welfare, important examination regulations and facilities for learning.

Student who will use the handbook will find it informative, educative and as a guide towards becoming successful graduates from the Department, who are useful to themselves and the entire society.

Dr. Mrs. B.E. Uwameiye
Ag. Head of Department

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**DEPARTMENT OF VOCATIONAL AND TECHNICAL EDUCATION
AMBROSE ALLI UNIVERSITY
EKPOMA
LIST OF ACADEMIC STAFF**

S/N	NAME	QUALIFICATIONS	RANK
1.	Dr. Mrs. B.E. Uwameiye	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Reader/H.O.D
2.	Prof. Raymond Uwameiye	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Professor
3.	Prof. Mrs. E.E. Idialu	B.Sc (Ed.), M.Ed. Ph.D	Professor
4.	Dr. Mrs. J.A. Adavbiele	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Reader
5.	Dr. P.S.O. Uddin	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Reader
6.	Dr. V. O. Uwaifo	B.Sc(Ed.), MBA, M.Ed, Ph.D.	Reader
7.	Dr. Mrs. J.I. Oviawe	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Senior Lecturer
8.	Dr. Mrs. P. Aluyor	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Senior Lecturer
9.	Dr. F.E. Iserameiya	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Senior Lecturer
10.	Dr. C.J. Okosun	N.C.E, B.Sc (Ed.), M.Ed. Ph.D	Lecturer II
11.	Mrs. J. Abhulimen	N.C.E, B.Sc (Ed.), M.Ed.	Assistant Lecturer
12.	Mrs. J.E Agege	B.Sc(Ed.), M.Ed.	Asst. Lecturer

List of Lecturers teaching Education Courses

S/N	NAME	QUALIFICATIONS	RANK
1	Prof. L.I. Aguele	B.Sc (Ed), M.Ed., Ph.D	Professor/Dean
2	Prof. R. O.A Aluede	B.Ed., M.Ed., Ph.D	Professor
3	Prof. O. Aluede	B.Ed., M.Ed., Ph.D	Professor
4	Prof. E.O. Imhamlahmi	B.Ed., M.Ed., Ph.D	Professor
5	Prof. (Mrs.). E. O. Omoregie	B.Ed. M.Ed., Ph.D.	Professor/Dean
6	Prof. A. I. Ojugo	B.Ed., M.Ed., Ph.D	Professor
7	Prof. Mrs. O.C. Otote	B.Ed., M.Ed., Ph.D.	Professor
8	Prof. I. O. C. Adomeh	B.Ed., M.Ed., Ph.D.	Professor
9	Prof. Mrs. J. Afen-Akpaيدا	B.Ed., M.Ed., Ph.D.	Professor
10	Dr. B.O. Jimoh	B.Ed., M.Ed., Ph.D	Reader
11	Dr. (Mrs.) J.O. Eimuhi	B.Sc., M.Ed., Ph.D	Reader
12	Dr. Mrs. P.E. Onolemhenmhen	B.Sc. Ed, M.Sc., Ph.D	Senior Lecturer

List of Lecturers from other Servicing Departments

S/N	NAME	QUALIFICATIONS	RANK
1	Prof. F. U. Igene	NCE, B.Sc., M.Sc., Ph.D	Professor
2	Dr. F. O. Ogedegbe	NCE, B.Sc. ED, M.Sc. Ph.D	Senior Lecturer
3	Prof. A. U. Omoregie	B.Sc., M.Sc., Ph.D	Professor
4	Mr. P.I Okoh	B. Agric., PGDE, M.Sc	Lecturer II
5	Dr. M. Okoruwa	B.Sc., M.Sc. Ph.D.	Senior Lecturer
6	Dr. S.O. Dania	B. Agric., M.Sc. Ph.D.	Senior Lecturer
7	Dr. E.E Ikheloa	B.Sc., M.Sc. Ph.D.	Senior Lecturer

8	Dr. C.E. Isidahomen	B.Sc., M.Sc. Ph.D.	Senior Lecturer
9	Dr. M.E. Alleh	B.Sc., M.Sc. Ph.D.	Senior Lecturer
10	Mr. E.T.E. Ehebha	B.Sc., M.Sc. Ph.D.	Lecturer I
11	Mr. C.A Igene	B.Sc., M.Sc. Ph.D.	Asst. Lecturer
12	Mr. O.O. Egaghara	B.Sc., M.Sc. Ph.D.	Lecturer 1
13	Dr. V.J. Adavbiele	B.Sc., M.Sc. Ph.D.	Lecturer 1
14	Mr. E.S Oseghale	B.Sc., M.Sc	Lecturer II
15	Dr. S.O. Aigbe	B.Sc., M.Sc. Ph.D	Reader
16	Mr. Aikpitayi	B.Sc., M.Sc	Asst. Lecturer
17	Mr. D.O Obasoyo	B.Sc., M.Sc	Lecturer II
18	Prof. F.O. Ikpotoke	B.Sc., M.Sc. Ph.D.	Professor
19	Dr. S.E. Nnebe	B.Sc., M.Sc. Ph.D.	Reader
20	Mr. P.A. Aliga	B.Sc., M.Sc.	Lecturer. I
21	Dr. Mrs A. Odia	B.Sc., M.Sc. Ph.D.	Reader
22	Prof. S. Okecha	B.Sc., M.Sc. Ph.D	Professor
23	Dr. . E.E. Egbon	B.Sc., M.Sc. Ph.D	Reader
24	Dr. O.K. Ize-Iyamyu	B.Sc., M.Sc. Ph.D	Reader
25	Dr. E.O. Jatto	B.Sc., M.Sc. Ph.D	Senior Lecturer
26	Dr. S. Ehika	B.Sc., M.Sc. Ph.D	Senior Lecturer
27	Dr. O.E. Akhirevlulu	B.Sc., M.Sc. Ph.D	Senior Lecturer
28	Dr. Ogidi-Odia Efosa	B.Sc., M.Sc., Ph.D	Lecturer I
29	Dr. Osagie Ekhaton	B.Sc., M.Sc., Ph.D	Senior Lecturer

HISTORY OF THE PROGRAMME/SUB-DISCIPLINE/DISCIPLINE

Ambrose Alli University, Ekpoma was established by the First Executive Governor of the Defunct Bendel State, Professor Ambrose Folorunsho Alli in 1982. The University was opened for full academic operations on January 15, 1982. Under a Collegiate structure, the College of Education was one of the eight Colleges which started in the University. The College of Education started with 13 lecturers and Professor J. U. Aisiku was the first Rector of the College. The College of Education started with 44 students, who matriculated along with other students on February 6, 1982. In 1983, the then Executive Governor of Bendel State, Dr. S.O. Ogbemudia instituted a visitation panel on the University. The outcome of the report led to the rationalization of three Colleges, namely, the College of Education, College of Agriculture and College of Medical Sciences by Gen. Jeremiah T. Useni, Military Administrator of the State.

During the 1985/1986 academic session, the Faculty of Education was reopened in the University and was housed in the then College of Education, Abraka. That led to the transformation of the College of Education, Abraka (an N. C. E awarding institution) to become a campus of Bendel State University. Following the creation of additional States in the Federation of Nigeria, the then Faculty of Education of the University which was situated at Abraka was acquired by the Delta State Government and upgraded to a full University.

Following the acquisition of the Abraka campus of the then Bendel State University by the Delta State Government, it became obvious that the Ekpoma main campus of the University which was renamed Edo State University, Ekpoma was without a Faculty of Education. Thus, during the 1992/93 academic session, Senate of the University recommended the establishment

of the Faculty of Education for the University, which was established and had Professor J. U. Aisiku as the Dean.

The Faculty operated then under three Departments: Department of Educational Foundations, Department of Curriculum and Instruction and the Department of Vocational and Technical Education. In 1996/1997 session Department of Physical and Health Education was separated from the Department of Vocational and Technical Education as they were one Department before then. As at that time, the Department offered five degrees options namely, Agriculture Education, Business Education (Accounting and Secretary Education), Home Economics Education and Industrial Technical Education with options in Building, Electrical/Electronics and Mechanical Technology.

In 1997/1998 session, because of lack of manpower, accommodation facilities and other modalities for accreditation purpose, Industrial Technical Education curriculum approved by the Senate was suspended and a new programme the 'Technology Education' was put in place instead of the 'Industrial Technical Education', which is being offered till date. It is now that plans are under way to restore back the Industrial Technical Education curriculum approved by Senate of the University to replace the Technology Education.

In the history of the Department, the following persons have emerged as Heads in the Department: Prof. A. Momodu, Prof. M. O. Omo-Ojugo, Prof. Don. Omoike, Dr. Osemota, Dr. A. I. Iseguan, Prof. E. O. Imhanlahimi, Prof. Mrs. H. Efoghe, Prof. Mrs. E. E. Idialu, Prof. R. Uwameiye, Dr. Mrs. J. A. Adavbiele, Dr. Rev. P. S. O. Uddin, Late Dr. G. O. Agbonghale, and Dr. Mrs. B. E. Uwameiye.

The Department has just in 2015/2016 session started the higher Degree programme in M.Ed, MEd/M.Phil and Ph.D in Home Economics Education and Technical Education.

GUIDELINES OF THE COURSE SYSTEM

1. CATEGORIZATION OF COURSES:

1. **CORE COURSES (C):** These are courses within the discipline which must be taken and passed.
2. **REQUIRED COURESE (R):** These are courses outside the discipline which must be taken on the advice of the Department and passed.
3. **GENERAL STUDIES COURSES (G):** These are courses of the General Studies Programme which must be taken and passed.
4. **ELECTIVE COURSES (E):** These are courses within or outside the discipline that are to be taken to meet the requirements of a particular degree programme but need not be passed.
5. **PRE-REQUISITE COURSES:** Pre-requisite courses are courses the knowledge of which is needed prior to the taking of other specified courses. A student is deemed to have obtained this pre-requisite knowledge if he/she obtains a mark not less than 30% but will not be credited with any units in the course concerned except he/she scores a minimum of 40%. This particular clause is without prejudice to Faculty requirements.

II. COURSE ASSESSMENT:

- (a) Every course assessment must consist of continuous assessment
- (b) (between 20% and 40%) and course examination (between 60% and 80%).
- (c) The pass mark for every course assessment is 40%.
- (d) The grading system is as follows:

Scores	Grade	Grade Point
70-100	A	5
60-69	B	4
50-59	C	3
45-49	D	2
0-39	F	0

However, the university has started implementing NUC new grading system with effect from 2013/2014 session.

- (e) Students' results are to be prepared at the end of every session reflecting the units taken, the units passed (accumulated) and the semester's G.P.A.

- (f) At the end of every session students' results are prepared reflecting the units taken during the session, the units passed during the session, the cumulative Grade point average (CGPA), the courses failed for the session and the over-all result of proceeding, on probation or withdrawal from the (degree) programme.
- (g) At the end of the degree programme students' results are prepared reflecting total units registered in the programme, total units passed in the programme, CGPA. Courses failed and degree classification according to the following scheme.

CGPA	CLASS OF DEGREE
4.50-5.0	First Class
3.50-4.49	2 nd Class Upper
2.40-3.49	2 nd Class Lower
1.50-2.39	Third Class
Less Than 1.50	Fail

This is for the old students.

- (h) The CGPA for a student is determined in the following manner:
- (i) For each session the weighted grade point is obtained for each course as the product of the grade point and the units for the course.
- (ii) The total weighted grade points and the total units are obtained for each session.
- (iii) The sum of the total weighted grade points for all the sessions and the sum of the total units for all the sessions are calculated to give the cumulative weighted grade points and the cumulative units respectively.
- (ii) On dividing the cumulative weighted grade points by the cumulative units one obtains the cumulative Grade Point Average (CGPA).
- (h) As an example consider a student who takes seven courses in a session with the following details:

	Units (a)	Mark (B)	Grade (c)	Grade Point (d)	Weighted Grade Point (a) x (d)
Course 1	3	62	B	4	12
Course 2	3	51	C	3	9
Course 3	3	48	D	2	6
Course 4	2	33	F	0	0
Course 5	3	45	D	2	6
Course 6	2	52	C	3	6
Course 7	3	42	D	2	6
					45

- (ii) If a student has the following results over four sessions:

Then cumulative weighted grade point = 284
 Cumulative Units = 164
 Hence, CGPA = 1.73

The student will therefore come up in the third class degree classification.

- “ There is no reference in any course examination.
- “ There is no repeat in the course system. Therefore a student cannot re-register for a course already passed.
- “ A student must accumulate at least 30 units per level before graduation.
- “ There is no weighting of sessional GPA in the computation of CGPA.
- “ In the computation of the CGPA all courses taken in the session will be used, and therefore no course will be disregarded or discountenanced.

PROBATION:

1. A student who makes a CGPA of 1.50 or more at the end of the session will proceed to the next level of the degree programmed for which he is registered.
2. A student who makes a CGPA of less than 1.50 at the end of the session will be on probation for the following session to enable him/her improve on the CGPA. During that session he must register for the appropriate core-courses, required courses and GST courses which he/she has not passed, and any other courses for which he/she has the pre-requisites.
3. A student on probation during a session who makes a CGPA of less than 1.50 during that session must withdraw from the degree programme for which he is registered.
4. If the student changes to a new degree programme and obtains a CGPA of less than 1.50 in the new degree programme he/she will again be on probation. If however he obtains a CGPA of less than 1.50 a second time in the new degree programme he/she will be asked to withdraw from the University.

TRANSFER:

1. Every student seeking transfer from one degree programme to another must complete the necessary forms within the stipulated time.
2. All courses taken in the previous degree programme will be used for the computation of the CGPA for the new degree programme.
3. All regulations in respect of the new degree programme concerning core courses, required courses, etc., must be met before graduation.

DEGREE PROGRAMME: B.Sc. Agricultural Science Education

PHILOSOPHY AND OBJECTIVES

PHILOSOPHY

Agricultural Education is a vocational Education programme. These graduates are expected to acquire relevant competence to enhance commitment to the teaching profession and also to be self employed/reliance in absence of getting jobs. Graduates of this programme (Agricultural Education) can obtain employment at the Junior and Senior Secondary Schools, Technical Colleges, Colleges of Education, and Polytechnics.

OBJECTIVES

In Line with the above philosophy, the objectives of Agricultural Education Programme are, to:

1. Acquire knowledge of basic Agricultural Science concepts, principles theories laws and conception schemes
2. Enable students to acquire necessary teaching and practical skills and other aspects of teaching methodology of their subject;
3. Enable students to become effective classroom teachers;
4. Expose students to industrial science;
5. Enable students to develop the ethics of teaching as profession and become professional Agricultural Science Education
6. Develop positive values and attitudes for efficient discharge of their duty as students disseminate information in Agricultural Science and related areas
7. Develop positives values and attitudes for efficient discharge of their duty as te

(a)

(a)

Admission Requirements:

UTME CANDIDATES

A candidate must have at least five credit passes in Senior Secondary certificate Examination of NECO or its equivalent in not more than two sittings including English Language, Mathematics Agricultural Science and Biology, Chemistry, Economics and Physics

I. UME- Four-Year Programme

A candidate must have at least FIVE credit passes in Senior Secondary certificate Examination of NECO or its equivalent in not more than two sittings including English Language, Mathematics, Agricultural Science and Biology.

II. Direct Entry-Three –Year Programme

Candidate for Direct Entry must poses one of the following

- III. Passes at merit levels in at least two subjects in NCE with GCE O/L credit or its equivalent in three other subjects excluding teaching practical
- IV. Passes in two relevant subjects at A/L with GCE O/l credits or its equivalent in three other subjects at not more than two sittings.

Passes at merit level in a relevant Diploma programme of recognized University in addition to three O/L credit passes.

COURSE: Agricultural Science Education

LEVEL: 100

FIRST SEMESTER

COURSECODE	TITLE	UNITS/STATUS	
EDU 100	Introduction to Teaching Profession	2	C
EDU 101	History of Education	2	C
EDU 102	Developmental Psychology	2	C
GST101	Use of English and Library	4	C
GST102	Philosophy and Logic	2	C
CSC101	Introduction to Computer Science	2	C
CHM101	General Chemistry I	3	C
BIO 101	General Biology	4	C
AGR 101	Introduction to General Agriculture	<u>3</u>	<u>C</u>

TOTAL	24
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SECOND SEMESTER

EDU113	Introduction to Science and Technical Education	2	C
GST 111	Nigeria People and Culture	2	C
GST 112	History and Philosophy	2	C
CHM 112	General Chemistry II	3	C
BIO 111	General Biology II	4	C
PHY103	General Physics Laboratory	2	C
AGR 111	Farms Practical	<u>2</u>	<u>C</u>

TOTAL	<u>17</u>
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EDU 115	Introduction to Special Education	<u>2</u>	<u>E</u>
		2	

GRAND TOTAL	24+17 = 41
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LEVEL : 200

FIRST SEMESTER

COURSECODE	TITLE	UNITS/STATUS	
EDU 201	Philosophy of Education	2	C
EDU 202	Curriculum and Instruction	2	C
ENT201	Entrepreneurship Education 1	2	C
AED201	Climatology and Biogeography	2	C
AED202	Anatomy and Physiology of Farm Animal	2	C
AGR203	Introduction to Crop Protection	<u>3</u>	<u>C</u>
TOTAL		<u>13</u>	

FOR DIRECT ENTRY (They add these)

GST 101	Use of English and Library	4	C
GST102	Philosophy and Logic	2	C
CSC 101	Introduction to Computer Science	<u>2</u>	<u>C</u>
	TOTAL	8	
	Grand Total	21	

SECOND SEMESTER

EDU211	Subject Methodology	3	C
EDU212	Measurement and Evaluation	3	C
EDU213	Sociology of Education	2	C
EDU 312	Research Method and Data Processing	3	C
GST222	Peace and Conflict Resolution	2	C
ENT211	Entrepreneurship Education II	2	C
AED211	Principle of Crop Production	2	C

AED212	Principles of Soil Science	2	C
AED214	Introductory Agriculture Biochemistry	2	C
AED215	Principle of Animal Production	<u>2</u>	C
TOTAL			<u>23</u>

FOR DIRECT ENTRY (They add these)

GST 111	Nigerian People and Culture	2	C
GST112	History and Philosophy of Science	2	C
		<u>4</u>	
Grand Total		27	

Normal Entry	13+23 =	36
Direct Entry	21+27 =	48

LEVEL: 300

FIRST SEMESTER

COURSECODE	TITLE	UNITS/STATUS	
EDU 300	Teaching Practice	3	C
EDU301	Education Administration and Planning	2	C
EDU302	Educational Technology	3	C
AGR205	Crop Anatomy, Taxonomy and Physiology	3	C
AGR301	Non-Ruminant Animal Production	3	C
AGR304	Introduction to Pedology and Physics	3	C
AGR307	Ruminant Animal Production	3	C
VTE 301	In Nigeria and other countries	2	C
VTE308	Computer Technology	<u>2</u>	C
TOTAL		24	

SECOND SEMESTER

VTE311	Student Industrial Work Experience Scheme	<u>15</u>	C
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GRAND TOTAL	24+15 =	39
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LEVEL : 400

FIRST SEMESTER

COURSECODE	TITLE	UNITS/STATUS	
VTE400	Research Project/Seminar	6	C
EDU401	Practical Teaching	3	C
EDU402	Guidance and Counselling	2	C
EDU403	Continuous Assessment	2	C
CRP501	Crop Husbandry	4	C
AGE506	Admin. Sup. In Extension 1	<u>2</u>	C
TOTAL		19	

SECOND SEMESTER

EDU311	Educational Psychology	2	C
EDU411	Organization of Primary and Secondary Education	2	C
EDU413	Comparative Education	2	C
AGR311	Permanent Crop Production	2	C
AGR312	Soil Chemistry and Microbiology	3	C
AGR 314	Extension Teaching Learning Process &Method	2	C
AGR315	Agriculture Biochemistry and Methods	3	C
AGE511	Agric. Business Management and Accounting	<u>2</u>	<u>C</u>
TOTAL		18	

GRAND TOTAL**19+ 18 = 37****COURSE DESCRIPTION****COURSE DESCRIPTION: B.Sc Agricultural Science Education****LEVEL: 100****AGR 101 - INTRODUCTION TO GENERAL AGRICULTURE (3UNITS)**

The definition of agriculture, world production and food supply, history, scope of Characteristics features of tropical Agriculture and how they affect production. Land use and tenure in the production distribution and Utilization of agricultural products. Measures of improving Nigerian Agriculture. Climatic, adaptic and social factors in relation to crop production and distribution and significance of farm animals.

Basic, principles of animal farming, place of forestry, fish and farming and wild life in Agriculture. Career opportunities in Agriculture.

BIO101 - General Biology (4Units)

Cell structure and organization, functions of cellular organelles. Diversity, characteristics and evolution elements of ecology and types of habitat.

CHEM101 - General Chemistry 1 (3Units)

Atoms, Molecules: Atomic structure, the development of atomic, theory, Electronic configuration. Chemistry bonding: Electrovalent, Covalent, dative and multiple bonding, chemical equations stoichiometry. Gases: the Gas Laws. The idea (or perfect) gas equations: Real gases. Critical Constants. Chemical Equilibria. Equilibrium constants) KC and Relationship the position or equilibrium. Le Chatellier principle. Solubility products; products; partition law; colligative properties; Acids and Bases.

Thermodynamics: various types of thermodynamic system. First Law of thermodynamics. Factors effecting rates of reacting Electrolytes and non-electrolytes. Laws of electrolysis and their applications.

CSC101 - Introduction to Computer Science (2Units)

History of computer, functional components Characteristics of a Computer.

Understanding Disk Operating System (DOS) and it file structure environment. Working with dialogue boxes. The use of programme and file managers. Menus and their uses. Problems

solving flow charts Algorithms Computers, programming statement, symbolic name arrays, subscripts, expressions and control statements, data representative, the decimal system, the binary system, numbers complement, introduction to BASIC or FORTRAN programming language, Computer Applications.

CHM 102 - PRACTICAL CHEMISTRY (2Units)

Acid –base titration, Redox titration. Synthesis and preparations of inorganic compounds. Analysis of selected anions and cations. Preparations and qualitative analysis of organic compounds pH measurement.

PHY 103 - GENERAL PHYSICS LAB (2Units)

Mechanics, optics, electricity, magnetism.

SECOND SEMESTER

BIO111 - General Biology II (4Units)

A generalized survey of plant and animal Kingdom based mainly on study of similarities and differences in the external features, ecological adaption of these forms.

AGR111 - FARM PRACTICAL (2Units)

Plot are allowed to plant students them to have field experience on agronomic practices on various crops. Please see the Faculty of Natural Science handbook for other 100 levels courses descriptions.

CHM112 -General Chemistry II(3Units)

Periodic classification of the elements; the periodic table groups and periods. Relationship in the table of main group elements the periodic law, anomalies in the periodic table. Valence forces, structure of solids. Chemistry of selected metals and non metals/organic Chemistry definition history classification of organic compounds non-menclature homologous series functional group (i.e alkane, alkanes alkynes alcohols phenoesthoids aldehydes ketone carbonxylic acids ether esters amide nitrile haloalkens sulphoxides and sulphones/basic stereochemistry electronic theory in organic Chemistry, saturated hydrocarbons unsaturated hydrocarbons.

200 LEVEL

AED 201 - CLIMATHOLOGY AND BIOGEOGRAPHY (2Units)

The principles, aim and scope of climatology and biogeography. The elements of control of climate and weather and the dynamics of the earth atmosphere systems, radiation, and heating of the atmosphere systems, atmosphere moisture, the dynamics of pressure and wind systems.

Consideration and precipitation processes. Seasonal variations in temperature, day length, radiation, rainfall and evaporation equipment and maintenance of standard meteorological stations. The tropical climate relation between agriculture and climate with reference to crops, livestock, livestock irrigation pests and diseases.

AED 202 - ANATOMY AND PHYSIOLOGY OF FARM ANIMAL (2Units)

Parts of beef and dairy cattle, sheep, goats, pigs, rabbits and poultry. Fundamental of cell biology, anatomy and physiology of animal tissues, nervous systems skeletal systems, muscle,

bone circulated system, reproductive, digestive, special sense and other systems of farm animals physiology functions of animal homeostasis, nutrition and reproduction. Endocrinology. The blood and circulation lactation, milk let down and egg production. Water balance.

AGR 203 - INTRODUCTION TO CROP PROTECTION (3Units)

Introduction to the structure, life history classification and importance of insect, nematodes, bacteria and virus.

AGR 205 - ANATOMY, TAXONOMY AND PHYSIOLOGY (3 Units)

Parts of the cell types, development of cell types. Introduction to plant taxonomy. Characteristics, distribution, economic importance and local examination examples of Leguminosae Gramineae, Solanaceae composite Discoraceae, mutaceae, comparative anatomy of major plant organs, Enzymes. Photosynthesis and transpiration seed dormancy and pollination, respiration and energy utilization. Seed dormancy and germination, development of fruits water relatives in plant growth and development, mineral nutrition.

AED 211- PRINCIPLE OF CROP PRODUCTION (2Units)

Crop production and its development the principles, and problems and prospect of crop production. Importance of crop rotation, cultural practice water and soil conservation irrigation and drainage. General types and characteristics of arthropods, micro organism and pests affecting crops. Weed and their effects on crop production. Pest, disease and weed control. Basic mendelian genetics, principles of crop production, harvesting processing and storage.

AED 212-PRINCIPLES OF SOIL SCIENCE (2 Units)

History, scope and development of soil sciences branches of soil science and relationship with other sciences. Definition of soil, component, soil solids (inorganic, soil water, soil air and oil organism). Important physic-chemical properties of soil texture, structure, colour, porosity, tilth, PH, salt content and catio exchange capacity. Soil forming rocks and composition. Weathering of rocks and minerals. Factors of soil formation and soil forming process-soil profile horizon and differentiation. Introduction to soil survey and classification.

AED 214-INTRODUCTORY AGRICULTURAL BIOCHEMISTRY (2 Units)

Biochemistry of carbohydrates; lipids protein, enzymes and nucleic acids. Vitamins, Co-Enzyme and their functions, nature, classification and function of enzymes and hormones.

AED 215-PRINCIPLES OF ANIMAL PRODUCTION (2 Units)

Animal production and its development. The livestock industry – problems and prospects. Description of the breeds of cattle, sheep. Goats, pigs, poultry and rabbits. Systems livestock production. Feeding habit of farm animals. Principles of management of the different type of farm animals. The importance of fishes and wildlife of West Africa with emphasis on Nigeria species. The principles of fish and Wildlife species in Nigeria.

AGR 301-NON-RUMINANT ANIMAL PRODUCTION (3Units)

Management of breeding stock and young animals housing equipment Feeding principle of poultry rabbits economics health management of Stock, processing and marketing of poultry, pigs and rabbit. pre-request

AGR 304 - INTRODUCTION PEDOLOGY & PHYSICS (3UNITS)

Soil its origin and formation, soil morphological characteristics, soil components soil formation rock and soil survey. Soil mapping, soil classification. Properties and management of Nigeria soils. Classification of soil separates; soil texture, surface area of particles, aggregation. Soil structures and stability porosity, soil water relations, soil and hydro-logic cycle. Soil temperature and condition, soil erosion.

AGR 307 - RUMINANT ANIMAL PRODUCTION (3 Units)

Management of breeding stock, growing and young animals housing equipment and principles of cattle, sheep and goats production and management practices. Health management of ruminant animals. Processing and marketing of cattle sheep and goats Ruminant animal products.

AGR 311 - PERMANENT ANIMAL PRODUCTION (2 Units)

Origin distribution, soil and climatic requirement of some important crop such as cocoa, oil palm, rubber, coffee, coconut, mango, sugar-cane, banana, plantains, citrus, kola, cashew etc. production practices, improvements harvesting utilization, processing, storage and economic aspects of some elected permanent and perennial crops.

AGR 312 - SOIL CHEMISTRY AND MICRO-BIOLOGY (3 Units)

Chemistry composition of soils, soil fertility, conversion Units and calculations. Soil fertility evaluation. Silicate mineral chemistry, cation and anion exchange phenomena and base saturation. Soil reaction (active and reserve acidity, alkalinity, buffering capacity). Soil acidity and liming survey of microorganism in soil and their role in soils. The dynamic of N. P. and S. Association between microbes and plant.

AGR 314 - EXTENSION TEACHING, LEARNING PROCESS AND METHODS (2 Units)

Nature and elements of communication process. Principles of communication. Application of communication process of analyzing communication problems in extension. The meaning of the concept of learning, teaching and motivation. Steps and principles of teaching and learning extension teaching methods, preparation and use of teaching material and aids.

AGR 315 - AGRIC. BIO CHEMISTRY AND METHODS (3 Units)

The animal body composition and its feeds. Biochemistry of photosynthesis. Metabolism of Carbohydrates, lipids, proteins and nucleic acid chemistry and mode of action of enzymes and hormones. Chemistry methods and instrumentation and their application in the analysis of selected agricultural products.

CRP 501 - CROP HUSBANDRY (4 Units)

History, definition classification and importance of vegetables, field and plantation crops. Ecological distribution of crops in Nigeria. Varieties and adaptation of crop to the Nigerian environment. Types and system of crop production, production practice, harvesting, handling, processing, storage, marketing and utilization of vegetables and tropical crops. Method of plant propagation. Nursery system, disease and pests of vegetable and tropical crops. Fertilization, water maintaining ornamental trees, shrubs, perennials and fruits in the nursery home and packs.

AGE 506 - ADMIN. AND SUPERVISION IN EXTENSION I (2 Units)

Concepts, theory and guideline and principles of extension organization, management, and supervision roles and responsibilities of various levels of extension and other relevant staff recruitment, selection placement and supervision. Plan principles of morale and motivation. Implication of extension staff development and promotion. Creating conducive working environment, discipline, assessment of extension work accomplishment. Improving Nigeria extension services.

AGE 511 - AGRIC. BUSINESS MANAGEMENT AND FINANCE (2 Units)

The scope of agricultural business and management, types of agricultural business management and organization, enterprise selection, production planning public policies affecting agriculture business, farm growth, organization of large scale farms of agricultural processing marketing management principles of agricultural finance, principle of farm credit agencies and government credit policy and approaches to efficient management, farm accounting, inventory, balance sheet, cashbook analysis.

COURSE: B.SC. HOME ECONOMICS EDUCATION**Philosophy**

The programme is designed to prepare students to undertake a professional teacher education course by equipping them with the relevant knowledge and skills necessary for the pursuit of a University degree in Home Economics Education as a teacher and/or practitioner.

(b) Objectives

The objectives of Home Economics Education programme include:

- i to expose students to skills in Home Education towards teaching at Levels of Junior Secondary, Senior Secondary, and tertiary levels of education.
- ii to equip students with the intellectual and processional skills and competences required for self-reliance.
- iii to inculcate into students the necessity to strengthen family life through improving personal, family and community living.

ADMISSION REQUIREMENTS

- i Candidate for the programme must have Senior Secondary School Certificates: SSCE, NECO, NABTEB, GCE or other equivalent minimum of 5 subjects passed at credit level at not more than 2 sittings. The subjects must include English Language, Biology or any other science subjects, plus at least a pass in Mathematics.
- ii T.C.II with relevant qualifications will also be considered for admission.

Direct Entry Requirements

Candidates must possess one of the following:

- i Passes at merit level in at least two subjects in the N.C.E with SSCE or G.C.E O/L credit passes or its equivalence in five subjects, at not more than two sittings.
- ii Passes in two relevant subjects at A/L with G.C.E., O/L credit or its equivalent in five relevant subjects, at not more than two sittings.

- iii Passes at merit level in a relevant Diploma programme of Ambrose Alli University of any other recognized by the Senate, in addition to UME requirements.

COURSE: B.Sc. Home Economics Education

100 Level

FIRST SEMESTER

COURSE CODE	TITLE	UNITS/STATUS	
EDU 100	Introduction to teaching profession	2	C
EDU101	History of education	2	C
EDU 102	Developmental psychology	2	C
GST 101	Use of English and library	4	C
GST 102	Philosophy and logic	2	C
CSC 101	Introduction to computer science	2	C
HEC 101	Introduction to Home Economics	2	C
HEC 102	Introduction to clothing and textile	2	C
HEC 103	Introduction to Nutrition	2	C
CHM 101	General chemistry 1	3	C

SECOND SEMESTER

EDU 113	Introduction to Science and technology	2	C
GST 111	Nigerian people and culture	2	C
GST 112	History and philosophy of education	2	C
HEC 111	Applied Home Economics	3	C
HEC 112	Principles of food preparation	2	C
HEC 113	Clothing techniques	2	C
HEC 114	Meal management and service	2	C
HEC 115	Principles of Home management	2	C

ELECTIVE

EDU 115	Introduction to Special Education	2	E
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200 Level

FIRST SEMESTER

EDU 201	Philosophy of Education	2	C
EDU 202	Curriculum and Instruction	2	C
ENT 201	Entrepreneurship Education 1	2	C
HEC 201	Work Simplification and Household Equipment	2	C
HEC 202	Recipe Development and Testing	2	C
HEC 203	Pattern Drafting and Alteration	2	C
HEC 204	Principles of Nutrition	2	C
CHM 203	Organic Chemistry	3	C

FOR DIRECT ENTRY (They add these)

GST 101	Use of English and Library	4	C
GST 102	Philosophy and Logic	2	C
CSC 101	Introduction to Computer Science	2	C

SECOND SEMESTER

EDU 211	Subject Methodology	3	C	
EDU 212	Measurement and Evaluation	3	C	
EDU 213	Sociology and Education	2	C	
EDU 312	Research Method and Data Processing		3	C
GST 222	Peace and Conflict Resolution		2	C
ENT 211	Entrepreneurship Education II		2	C
HEC 211	Clothing Design by Flat Pattern		2	C

FOR DIRECT ENTRY (They add these)

GST 111	Nigerian People and Culture	2	C
GST 112	History and Philosophy of Science	2	C

300 Level**FIRST SEMESTER**

EDU 300	Teaching Practice	3	C
EDU 301	Educational Administration and Planning	2	C
EDU 302	Educational Technology	2	C
HEC 301	Food Preparation	2	C
HEC 302	Basic clothing Construction	3	C
HEC 303	Nutrition in Health and Diseases	2	C
HEC 304	Consumer Education	2	C
VTE 301	VTE in Nigeria and Other Countries	2	C
VTE 308	Computer Technology	2	C

SECOND SEMESTER

VTE 311	Student Industrial Work Experience Scheme	15	C
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400 Level**FIRST SEMESTER**

VTE 400	Research Project/Seminar	6	C	
EDU 401	Practical Teaching	3	C	
EDU 402	Guidance and Counseling	2	C	
EDU 403	Continuous Assessment		2	C
HEC 401	Advanced Clothing Construction		3	C
HEC 402	Clothing Maintenance	2	C	
HEC 403	Community Nutrition	2	C	
HEC 404	Home Management Residence		2	C

SECOND SEMESTER

EDU 311	Educational Psychology		2	C
EDU 411	Organization of Primary and Secondary Education	2	C	
EDU 413	Comparative Education		2	C
HEC 411	Family Relationship	2	C	
HEC 412	Housing and Interior Decoration		2	C
HEC 413	Needs of Special People		2	C
HEC 414	Extension Programme in Home Economics	2	C	

Course Description –B.Sc. Home Economics Education

FIRST SEMESTER

100 LEVEL

HEC 101; Introduction to Home Economics [2Units]

Definition and origin of home economics profession, department of home Economics in Nigeria. Historical, Sociological Philosophical and economics foundation of Home Economics. Home Economics as a multi-disciplinary subject, content, scope and object yes, concepts, principles and terminology, career opportunities in home economics. The educational value of home economics and its contributions in the developing country like Nigeria.

HEC 102: Introduction to Textiles and Clothing (2 Units)

Physical and chemical characteristics of various types of fibres (natural and synthetic) fabrics and finishes, selection, used and care for home use and appeared. (Study of clothing needs based on physical: aesthetic, economics, cultural and social psychological factors, basic clothing construction methods, equipment measurement dest of fabric abrasion, elongation colour fastness to perspiration crook, etc.

HEC 103 Introduction to Nutrition (2 Units)

Definition and concept of nutrition. Food and food nutrients, sources, functions and deficiency diseases. Dietary requirements of various groups with regards to age, sex health and activity.

SECOND SEMESTER

100 LEVEL

HEC 111 Applied Home Economics (food, clothing, textiles and home management) (3 Units)

Definition of foods, reason for cooking food, principles underlying and advantages and disadvantages of these methods, food hygiene, simple food storage and preservation methods in the home. Practical training in arts technique such as weaving, knitting and crocheting.

Design and fashion in the home embroidery, tie and dye constructing garment for toddlers of pre-scholars. Clothing renovation and patch work. Make an album showing different machines, principles involved in choice, use and care of household equipment modern and traditional. Methods of operation, maintenance of the equipment and their application in terms of their structure. Functions of each equipment, visit museum and art galleries within the state compare ancient and modern household. Write a report on them supported by facts from the elderly people in the community.

HEC 112: Principle of Food Preparation 1 (2 Units)

Acquisition of basic catering skills in the preparation of breakfast, lunch and supper dishes, pastries (Sweet and savoury), batters cake and yeast mixtures. Packed meals and echauffe Dishes. Egg, meal and fish dishes.

HEC 113: Clothing Technique (2 Units)

Practical selection of clothing and basic principles and processes of clothing construction for children pre-school and primary school children construction for children pre-school and primary school children. Simple pattern drafting.

HEC 114: Meal Management and Service (2 Units)

Basic management process applied to meal planning and preparation. Basic service styles in the home and institutions general meal service guideline. Table laying, table etiquette.

200 LEVEL

HEC 201: Work Simplification and Household Equipment (2 Units)

Management processes involving implementing, controlling, evaluation and modification of plans to improve the use of family resources use of money, time and energy in order to solve family problems thereby improving family living.

HEC 202: Recipe Development and Testing (2 Units)

Fundamental principles of food quality evaluation and development of standard, taste, flavor, shape size texture colour, and appearance, principles of recipe formulation and presentation. Acceptability trials, selection of participants analyses of results, emphasis on development and testing of more economical and nutritional foods from familiar and commonly used ingredients or new breeds (variety) of food stuffs.

HEC 203: Pattern Drafting and Alteration (2 Units)

Introduction to basic principles of design; technique used in development, alteration and styling of pattern through the use of drafting method. Fitting techniques project should include making up of at least one of the style drafted.

HEC 204: Principles of Nutrition (2 Units)

Food-its origin and importance nature of food structure and composition. Food groups and classification, source and functions. Inter-relationship of various nutrients to the body in health and malnutrition. Metabolic pattern, and trends of food consumption in relation to nutritional requirement for different ages and sex, fruits and vegetables, starches, flours and flour mixtures. Food habit formation cultural, social and psychological factors affecting food choice, preparation and consumption.

SECOND SEMESTER

200 LEVEL

HEC 211: Clothing Design by Flat Pattern Method (2 Units)

Creating of styles through the manipulation of parts, different necklines, sleeves, collars and skirts and effect a direction of lines, apply the principles of art to dress design. Clothes for special occasion. Utilization of special fabrics in creating fashion.

300 LEVEL

HEC 301: Principles of Food Preparation 11 (2 Units)

Application of principles of nutrition. Management and preparation of meals for special groups and occasions. Foods for different cultures in and outside Nigeria. Strategies for providing nutrient values and utilization of the traditional and untraditional coals, leavening agents.

HEC 302: Basic Clothing Construction (3 Units)

Basic construction technique with commercial patterns and fabrics selection; the application of these factors to consumer buying. Cultural and special aspects of clothing, taking into account the economics, climatic and special aspects of fabrics.

HEC 303: Nutrition in Health and Diseases (2 Units)

Nutrition during infancy, childhood, adolescence, pregnancy, location, and old age. Application of basic nutritional principles in the treatment of diseases such as kwashiorkor, marasmus, diabetes mellitus, obesity, hyperliodemia, ulcer; anemia and other metabolic diseases.

HEC 304: Consumer Education (2 Units)

Definition and principles of consumer education and analysis of economics forces affecting individuals and families as consumers of goods and services, creating carefulness of the rights and responsibilities of consumers in the market place, developing aids and techniques for making intelligent choice of goods and services advertisement and its influence on consumer choice of goods and services; government and non-government agencies that protect consumers print and non-print materials for solving professional and personal consumer problems.

VTE 301 VTE in Nigeria and other Countries (2 Units)

This course is designed to equip the students with concepts in Technical and Vocational Education. To achieve this objective the students should know (1) what is Technical and Vocational Education (2) The rationale for Technical and Technical Education, (3) policy and policy making for VTE (4) Organization Concepts and (5) Decision Making Process for VTE. (2) The Role of the VTE Teacher.

VTE 308: COMPUTER TECHNOLOGY (2 UNITS)

Computer utilization for instructional purposes. Using computer to assist in teaching tutorial, drill and practice. Computer software-important parts of the computer system, the opening system, topics of programmes which affect the operating system. Modes of operation and the electrical language used by the computer (code).

Cataloging the software. Compare hardware, identification of hardware. Description, function and types of input and output devices. The control processing and storage Units. Basic components and functions of a computer.

SECOND SEMESTER

VTE 311 Student Industrial Work Experience Scheme (SIWES) 15 Units

SIWES is a six months programme in which students are placed in industries or business establishment for the purpose of training on the job.

400 LEVEL

HEC 401: Advanced Clothing Construction (3 Units)

Study of advanced construction of clothes and techniques of handling new fabrics, the use of ready-made (commercial) patterns. Alteration principles; selection of colour and lines as related to individual and family needs.

HEC 402: Clothing Maintenance (2 Units)

Laundry apparatus and equipment; Laundry process (general and for specific natural and man-made fabrics) diabetic and stain removal agents, e.g. water soaps, bleaches, etc clothing care upholstery and household linen. Repairs, storage, etc.

HEC 403: Community Nutrition (2 Units)

Definition of community nutrition. Role of the community nutritionist. Determination and assessment of nutritional status of a community nutritional problems associated with urbanization, causes and solutions. Nation and international programmes aimed at improving nutrition of communities, natural and international agencies, concerned with food and nutrition in the community and their roles. Nutritional surveillance nutritional intervention programmes. Food registration, standards, and laws.

HEC 404: Home Management Residence (2 Units)

Practical demonstration of principles of home management under supervision students-live in and apply time, energy and money in maintenance the house. Run all the activities that go on in normal home. Each student must take turn at all duties so that work and experience gained in shared evenly. A period of 6-10 weeks in management techniques at family life cycle stages and socio-economics level.

Work covered should include: checking equipment and furnishing on errical, planning and cooking meals, shopping and keeping accounts, planning a routine of work for the house work and sharing duties, attending to personal and house laundry, checking equipment and furnishing when leaving the flat.

SECOND SEMSETER

HEC 411: Family Relationship (2 Units)

The family structure in Nigerian society and other culture interfamily relationship (the key terms and concepts of marriage and family. Family development and relationship; courtship marriage – forms advantages and disadvantages of each. Modes of divorce. Causes and solution.

Family planning-need of a family planning services. Methods; advantages and disadvantages of each population explosion, control and recommendation

HEC 412: Housing and Interior Decoration (2 Units)

Definition of interior decoration and furniture. The different part of the house and the furniture found in each room. Colour wheel. Basic principles of selection of family for home decoration. Construction of articles, furniture home decoration including craft and design.

HEC 413: Needs of Special People (2 Units)

Needs of special people. The handicapped, minority groups, community service to a selected person in need in helping them become self sufficient.

HEC 414: Extension Programme in Home Economics (2 Units)

Protein systems in food, sugar in food technology, pectic substance, plants gums and their uses in food; non enzymes browning, role of liquids in food lipid oxidation and lipid autoxidation in food system; carotenoids in food systems terpenes, essential oils and chemistry of food orders, phenolic compound respiration, chlorophyll and photosynthesis biochemical activity of micro-organisms in foods.

B.Sc. TECHNOLOGY EDUCATION PROGRAMME**1. Programme Title: B.Sc. Technical Education**

- 2. Philosophy:** There is ever increasing demand for highly qualified teachers in Nigeria. The aim of the faculty is therefore to produce academically and professionally qualified teachers and administrators for the post-primary, post-secondary institutions and other education sectors.

The department shall prepare students for the award of B.Sc. Technical Education, Degree of this University. The programme shall acquaint the students with basic theoretical and practical tools and techniques required for their future professional work in teaching and industries.

- 3. Objectives:** The programmes are designed to achieve the following objectives:

- i. To produce graduates who can teach effectively at the senior secondary schools and tertiary institutions.
- ii. To equip graduates to pursue work in education
- iii. To enable them to fit into administrative cadre in government and in institutions of higher learning.
- iv. To acquire skills and competence required for self-reliance and self-employment if the need arises.

4. Admission Requirements:

In addition to the general University requirements, the following regulations shall apply to the admission of students into the Department.

A. UME – Four-Year Programme

A candidate applying for a Bachelor of Science in Technical Education must meet the Tertiary Matriculation Examination (UTME) or Joint Admission and Matriculation Board (JAMB) requirements and in addition possess any of the following:

- i. West African Senior Secondary School Certificate or NECO or NABTEB with least five credit passes in not more than two sittings. The credit passes must include Mathematics and English, Physics or Chemistry plus any other relevant subjects.
- ii. National Technical Certificate (NTC) with credit passes in five subjects in not more than two sittings which must include Mathematics, English, Physics or Chemistry.
- iii. City and Guilds Certificate (Intermediate) in Electrical, Mechanical, Engineering subjects plus Federal Craft Certificate (FCC) with at least “C” grade in five subjects which must include Mathematics and English.
- iv. Teachers’ Grade II Certificate (TCII) with credit or merit passes in at least five subjects which must include Mathematics, English and General Science.

B Direct Entry Requirement (3 Years Programme)

Candidates with any of the following qualifications will be admitted into three years programme, provided they satisfy the university basic entry requirements. In addition, candidates must possess any one of the following:

- i N.C.E. (Technical)
- ii National Diploma (N.D.) or Advanced National Technical Certificate (ANTC) in relevant areas.
- iii City and Guilds (Finals) or Engineering Technicians Part II Certificate
- iv N.C.E. with a minimum of two passes in Physics/Chemistry, Chemistry/Mathematics, Physics/Mathematics and a pass in Education.

PROGRAMME/SUB-DISCIPLINE/DISCIPLINE STRUCTURE TO INCLUDE PERIODIC OF FORMAL STUDIES IN THE UNIVERSITIES, INDUSTRIAL TRAINING, PLANNED VISIT AND PROJECTS

Period of formal studies is one academic year, which covers the first and second semester. Each semester takes fifteen weeks. Students of Technical Education Programme are required to go for Industrial Work Experience Scheme in relevant engineering firms. Such industrial attachment is arranged during the second semester of the 300 level.

Students register for 15 units of industrial work experience (SIWES). Planned visit or field trip are arranged for students at least once before graduation to enable them develop insight into the operation in industrial centres such as power generating station, oil refinery, assembly plants for computer, automobile, cement factory, radio and television stations. In order to make such field trip a learning activity, the lecturer usually set up the objectives before the trip takes place. Students construct practical projects in the various areas of Technical Education as well as engage in group independent investigation of research topics pertinent to technical education in Nigeria.

TECHNICAL EDUCATION PROGRAMME B. SC (Ed) TECHNICAL EDUCATION

FOUR YEAR PROGRAMME IN TECHNICAL EDUCATION

100 LEVEL

FIRST SEMESTER

COURSE CODE	TITLE	UNITS	STATUS
EDU 100	Introduction to Teaching Profession	2	C
EDU 101	History of Education	2	C
EDU 102	Development Psychology	2	C
GST 101	Use of English and Library	4	C
CSC 101	Introduction to Computer Science	2	C
MTH 101	Elementary Maths 1 Algebra and Trigonometry	3	C
TED 101	Introduction to Technical Drawing	2	C
TED 102	Introduction to Building Construction	2	C
TED 103	Introduction to Metalwork Technology	2	C

Total Units = 21 units

SECOND SEMESTER

COURSE CODE	TITLE	UNITS	STATUS
EDU 113	Introduction to Science and Technology Education	2	C
EDU 115	Introduction to Special Education	2	E
GST 111	Nigerian People and Culture	2	C
GST 112	History and Philosophy of Science	2	C
TED 111	Health and Industrial Safety	2	C
TED 112	Building Construction and Material I	2	C
TED 113	Introduction to Electrical Technology	2	C
TED 114	Technical Drawing	2	C
TED 115	Introduction to Automobile Technology	2	C

Total core courses = 16 units

Total electives courses = 02 units

SUMMARY

Total units of core courses for the session	=	37 units
Total units of electives courses for the session	=	02 units
Grand Total	=	39 units

200 LEVEL

FIRST SEMESTER

COURSE CODE	TITLE	UNITS	STATUS
EDU 201	Philosophy of Education	2	C
EDU 202	Curriculum and Instruction	2	C
ENT 201	Entrepreneurship Studies I	2	C
TED 201	Technical Drawing II	2	C
TED 202	Structural Mechanics & Strength of Materials	2	C
TED 203	Electricity Technology	2	C
Total		12	

FOR DIRECT ENTRY

GST 101	Use of English & Library	4	C
GST 102	Philosophy & Logic	2	C
CSC 101	Introduction to Computer Science	2	C
Total		8	
Grand Total		20	

SECOND SEMESTER

COURSE CODE	TITLE	UNITS	STATUS
EDU 211	Method of Teaching Industrial Technical Education	3	C
EDU 212	Measurement and Evolution	3	C
EDU 213	Sociology of Education	2	C
EDU 312	Research Methods and Data Processing	3	C
ENT 211	Entrepreneurship Studies II	2	C
GST 222	Peace and Conflict Resolution	2	C
TED 211	Woodwork Technology I	2	C
TED 212	Engine Details/Transmission System	2	C
TED 213	Telecommunication Principles	2	C
Total Units	=	21	

SUMMARY

Total units of core courses for the session	=	33 units
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FOR DIRECT ENTRY

GST 111	Use of English & Library	2	C
GST 112	Philosophy & Logic	<u>2</u>	C
Total		<u>4</u>	

First Semester	=	20
Second Semester	=	25
Grand Total	=	45

300 LEVEL**FIRST SEMESTER**

COURSE CODE	TITLE	UNITS	STATUS
EDU 300	Teaching Practice	3	C
EDU 301	Educational Administration & Planning	2	C
EDU 302	Educational Technology	2	C
VTE 301	Time Management	2	C
VTE 308	Computer Technology	2	C
TED 301	Technical Drawing III	2	C
TED 302	Woodwork Technology	2	C
TED 303	Reinforced Concrete Design	2	C
TED 304	Mechanical Workshop Practice	2	C
TED 305	Breaking Suspension and Steering System	2	C
TED 306	Electrical/Electronics Mechanics	2	C

Total Units = 23 units

SECOND SEMESTER

COURSE CODE	TITLE	UNITS	STATUS
VTE 311	Students Industrial Work Experience (SIWES)	15	C

Total Units = 15 units

Grand Total = 38 units

400LEVEL**FIRST SEMESTER**

COURSE CODE	TITLE	UNITS	STATUS
VTE 400	Project	6	C
EDU 401	Practical Teaching	3	C
EDU 402	Guidance and Counselling	2	C
EDU 403	Continuous Assessment	2	C
TED 401	Law of Contract	2	C

TED	402	Glass and Plastics in Building	2	C
TED	403	Automobile Technology	2	C
TED	404	Practical Project Design in Special Area	2	C
TED	405	Electronics Circuit Laboratory	2	C

Total Units = 23 units

SECOND SEMESTER

COURSE CODE	TITLE	UNITS	STATUS
EDU 411	Organization of Primary and Secondary School	2	C
EDU 413	Comparative Education	2	C
EDU 311	Educational Psychology	2	C
TED 411	Electronics Power Technology	2	C
TED 413	Improvisation of Laboratory/Workshop Equipment	2	C
TED 415	Fabrication Technology	2	C
TED 416	Building Drawing	2	C
TED 417	Wood Constructional Methods	2	C
TED 418	Engine Lubrication/Cooling System	2	C
TED 419	Electrical/Electronics Drafting	2	C

Total Units = 20 units

Grand Total = 43 units

COURSE DESCRIPTION FOR TECHNOLOGY EDUCATION

100 LEVEL

FIRST SEMESTER

TED 101: Introduction to Technical Drawing 1

Use of Technical Drawing as a graphic language, drawing equipment and instruments, type of drawing board practice, drawing papers, mounting paper on drawing board, layout, title block styles, constructions, areas of plane figures, and pictorial drawings.

TED 102: Introduction to Building Construction

General principles of building construction, building construction tools, site plans, site preparation, setting out and elementary surveying, foundation, Damp Proof Course (DPC), mass-concrete, walls and timbering to trenches. Detailing of elements of buildings of all types: residential, industrial, commercial.

TED 103: Introduction to Metal Work Technology

Metals and non-metals, preparation and extraction of metals, properties and characteristics of metals, ferrous and non-ferrous metals, uses of metals, basic hand tools in a metal shop. Cutting tools: high carbon, medium carbon and low carbon steel, as used for cutting; other cutting tools used in metal workshop: tunsen, cementits, etc; metal lathe: parts, uses and operation; tool holders and work holding devices; types of chuck and their uses. Drilling machine: parts, uses,

and operation; grinding machine: parts, uses, and operation; milling machines: types of milling machines, and their cutters; various types and uses of milling cutters; methods of controlling table movement, power machine types, uses, and operations; and shaping machine: types, uses and operation.

SECOND SEMESTER

TED 111: Industrial Safety

Concept of safety, definition of industrial safety, safety education, hazard to safety and types of hazards, hazardous manifestations, safety measures, first aid treatment, accident prevention and management, causes and nature of industrial accidents, and good house and workshop keeping.

TED 112: Building Construction and Material 1

Types of floors and roofs. Drainage, external works and landscaping. Building regulations, partitions, doors and windows, roofs, internal finishing and external renderings. Chimneys and flues, stairways, scaffoldings, plumbing arrangement system, electricity and gas services and external works.

TED 113: Introduction to Electrical/Electronics Technology

Method of producing electricity, measurement of voltage, current and resistance to D.C. circuit power, primary and secondary cells, Basic electric circuits analysis, network theories, inductance and capacitance in disc circuits, resistance combination, magnetism and magnetic circuits, basic electrical/electronic tools and equipment, electronic symbols.

TED 114: Technical Drawing II

Solid geometry, advanced geometrical construction, conic sections, intersection of surfaces, locus problems, principle of tangency, pictorial and perspective projections, isometric scales and isometric projections, oblique projections, freehand sketching of solid, introduction to orthographic drawings.

TED 115: Introduction to Automobile Technology

Introductory explanation on the function of a car, its types and uses; basic discussion on the parts of an automobile:

- The Engine, its parts and materials for construction
- The clutch system
- The gearbox system
- Suspension, steering and lubricating system
- The electrical parts of the vehicle and its function.

200 LEVEL

FIRST SEMESTER

TED 201: Technical Drawing III

Interpretation of drawings, conversion of pictorial views and simple sectional views, parts drawings, preparation of working drawings, assembly drawing, and exploded views, cam construction, steel structural graphics, complex jigs and fixtures.

TED 202: Structural Mechanics and Strength of Materials

Stress and strain, plane framework, resolution and resolution forces, moment, roof truss and structural analysis.

TED 203: Electrical Technology

Basic A-C principles, sinusoidal rotation plotting and obtaining A-C values, Inductive, capacitive, resistive circuits A-C power; series and parallel A-C circuits; principle of electronic tubes – types, characteristics and applications; semi-conductor materials and conduction PN unction.

SECOND SEMESTER**TED 211: Woodwork Technology 1**

Botany, growth structure and identification of solid timber, its usages, market forms; introduction to woodwork tools and machines used in woodwork joints and joining; types of timber; sawing or conversion, seasoning, defects of timber, manufacture boards, woodwork joints, woodwork machines, machine preparation for timber, surface planer, saws, drills and shapes, etc; and their uses in simple projects.

TED 212: Engine Details/Transmission System

Understanding the functions of an automobile engine; Engine and its parts: engine block, top cylinders, valves, pistons, rings, connecting rods, flywheel, crankshaft, camshaft, etc. two stroke and four stroke spark ignition and compression ignition engine carburetor, its function, how the engine run the wheels; the transmission channels, purpose, types and uses of gearbox, clutch, propeller shaft and final drives, axle and road wheels. Automatic transmission system. Transmission without propeller shaft.

TED 213: Telecommunication Principle

Electromagnetic waves, method of propagation of waves, modulation and side bands; transmitter: two ways and four ways communication system and application.

300 LEVEL**FIRST SEMESTER****TED 301: Technical Drawing IV**

Orthographic projections in first and third angle projections, sectional views, auxiliary views and projections, advance conic auxiliary projections, lines in space and links mechanism.

TED 302: Woodwork Technology II

Iron-mongery with emphasis on carpentry and joinery and building construction contemporary and advanced methods of wood joinery modern techniques and machines work and decorative techniques.

TED 303: Reinforced Concrete Design

Introduction to reinforce concrete structures, water, aggregates, and cement; design and detailing of columns, reinforced concrete walls, bonds and limit state of cracking in building, concrete mixing, curing, and compacting of concrete. Technical report in building.

TED 304: Mechanical Workshop Practical

Forging and machining, application of the lathe machine in boring, screw cutting, V-threads, simple and double square threads including some threads, milling of flat surfaces.

TED 305: Braking, Suspension and Steering System

Brakes: function, uses and types – drum and disc brakes; hydraulic system. How the cylinder work together; working parts of drum and disc brakes; brake lining design and uses; parking brakes: types and operations; servo assisted system, hydro-pneumatic suspension; steering – what the mechanism has to do, types and operation; steering arrangement; toe in and toe out; steering ratio; camber and castor angles; power assisted steering.

TED 306: Electrical/Electronics Machine

Transformer principles, direct current motors and generators, armature winding, two and three phase connection, delta and star connection; synchronous generators and motors. Meter character.

VTE 301: VTE in Nigeria & other Countries

This course is designed to equip the students with the concepts of Technical Vocational Education and Training (TVET). To achieve this objective the students should understand: (i) the term Technical Vocational Education and Training; (ii) the rationale for TVET; (iii) policy and policy making for TVET; (iv) organization concepts; (v) decision making process for TVET; and (vi) the role of the TVET teacher.

VTE 308: Computer Technology

Computer utilization for instructional purposes. Using computer to assist in teaching tutorials, drills and practice. Computer software: important parts of the computer system, the operating system, topics of programmes which affect the operating system. Modes of operation and the electrical language used by the computer (code). Cataloging the software, compare hardware, identification of hardware; description, function and types of input and output devices. The control processing and storage units. Basic components and function of a computer.

SECOND SEMESTER

VTE 311: Students' Industrial Work Experience (SIWES)

Supervised learning activity consisting of practical work experience in industry. Acquisition of technical competence in latest and emerging technologies in business and industries. Students will be provided with opportunities to acquire up-to-date information on latest concepts, procedures and equipment used in their areas of specialization.

400 LEVEL

FIRST SEMESTER

VTE 400 Research Project

Students are expected to carry out independent studies in their own field of specialization. Such studies are expected to focus on finding solutions to the problems of teaching and learning.

TED 401: Construction Technology

Roof: functional requirements; factors that determine the most appropriate form of a roof structure in a given building; types of roof. Ceiling: major parts of a ceiling, steps to be followed in constructing a ceiling using asbestos sheets, Scaffolds: basic requirements, types, and procedures or steps for assembling a putlog scaffold. Stairs: building regulations that apply to stairs, terms used in stairs, types of stairs. Formwork: requirements of a good formwork, characteristics of a good formwork, considerations that should guide the design of formwork, types of formwork.

TED 402: Glass and Plastics in Building

Technology and manufacture, utilization in building, performance in building, maintenance problems. A critique of glass as external curtain, maintenance aspects and problems.

TED 403: Automobile Electricity

The ignition system, batteries types and its uses. Wet and dry cell batteries. Generators, dynamo and alternators: how they work. Headlamps, horn, windscreen wipers, side and rear lights. Dials and light to help the driver speedometer, millage recorder, tachometer, water temperature gauge, fuel gauge, oil pressure, temperature gauge, etc. refrigeration and air-conditioning.

TED 404: Practical Project Design in Area of Specialization

Students are expected to conduct or produce a practical project in their respective area of specialization. The project must be something of value and not a model and would give students the opportunity to exhibit theoretical and practical experience acquired in the course of the programme.

TED 405: Electrical/Electronics Circuit Laboratory

Radio receiver circuit, troubleshooting procedures and alignment. Troubleshooting turn-tables, cassette decks, installation of car radio sets, etc, resistors in service, parallel and series parallel, Ohms law verification, characteristics of indicators and capacitors, RC, RL, and RCL. Connection and determination of impedance diode and transistor characteristics.

SECOND SEMESTER**TED 411: Electronics Power Technology**

Analogue circuit-special diodes and their applications, Zener diodes, tunnel amplifier analysis and applications. Active filters, wave form shaping, power supply specification, single and three phase rectifiers circuits. Regulated power supply, 10 regulators, voltage multiplication. Digital techniques, binary system, basic circuits, flip flops. Counters and registers. Microprocessor. Simple A/D and D/A converters, comparators digital circuit fabrication.

TED 413: Improvisation of Laboratory and Workshop Equipment

Detailed discussion of materials and disposable wares that are found and used in Technical laboratories and workshops in secondary/technical schools. Methods and processes for designing/fabricating these materials locally.

TED 415: Fabrication Technology

Fabrication methods, casting and pattern design, Gorging and extrusion.

TED 416: Building Drawing

Building drawing instruments, building papers, building pencils work, tracing paper, ink, tracing pens and tracing itself; site plan; floor pal; building conventional and designs, foundation and basement; sectional details; elevations – front, sides, rear of bungalows; schedule of floors and windows, storey and multi-storey buildings, perspective drawing of building and structural details.

TED 417: Woodwork Constructional Method

Technical terms used in woodwork, elementary cabinet making, contemporary furniture sizes and design, shaping and bending of timbers, moldings. A series of drawings showing details of carcase work and drawer making. Finishing – concepts and applications; power method of finishing; laboratory work on different finishes, modern design and finishing techniques in the reproduction of completed items, a group of individuals project is required.

TED 418: Engine Lubrication and Cooling System

Reasons for cooling an engine parts of a modern cooling system- a jacked, radiator pump, can, hoses, thermostat, pressure cap, etc. thermo siphon cooling, corrosion, water pump, air and water cooled engine need for lubrication in engines. Materials used for lubricating engine pump, oil filters and how they work, lubricants – choosing the right oil, viscosity of oil and corrosion reduction.

TED 419: Electrical/Electronics Drafting

Pictorial drawing of electrical and electronic symbols, elementary diagrams including printed circuits, block diagrams, electric wiring diagrams of houses and distribution circuits.

COURSE CODE EDUCATION COURSE TITLES AND DESCRIPTION

EDU 100: Introduction to Teaching Profession

Definition of profession and teaching. Basic requirement needed by any occupation. Teaching in Nigeria and the contemporary debate on whether it is a profession or an occupation. Steps taken to professionalize teaching i.e. the various educational reforms.

EDU 101: History of Education.

Historical Background and education in Nigeria, colonial educational policy, and post-independence educational reforms. The course will also ex-ray various commissions set up by Colonial govt. as well as post Independence govt. to reforms educational system. And also contemporary issues and policy formulation and implementation of country educational system.

EDU 102: Developmental Psychology.

Stages of child development. i.e. prenatal and post-natal development. Development during infancy, early childhood, later childhood, adolescent and adult stage. Characteristics of children at various stages of development and their implication for educational planning and implementation. Cognitive theories about these developments shall also be x-rayed.

EDU 113: Introduction to Science and Technology Education

Historical and philosophical base of science and technology education. The relevance of science and technology education in developing nations. Emphasis should be laid on the relevance of science and education in Nigerian aspiration towards technological development of science and technology in Nigeria will also be discussed

EDU 115: Introduction to Special Education.

Definition or meaning of special education. Categories and exceptionality in children. i.e. visual impaired mentally retarded, learning disability, ordinarily impaired or deafness, emotional will also examine methods of educating handicap children as well as gifted or talented children.

EDU 201 Philosophy of Education.

The course will examine the various philosophical principles affecting education. Philosophical theories like existentialism. Idealism and pragmatism will also be discussed. Great Philosophers like Plato, Socrates, Aristotle, e.t.c. Will also be discussed. Philosophy of pragmatism will be discussed extensively with emphasis on their impact on educational reform or development.

EDU 202 Curriculum and Instruction

Philosophical base of curriculum development will be examined. Curriculum process and theories, curriculum objectives, purpose of objectives, agencies involved in curriculum development as well as processes of curriculum evaluation will be discussed. Furthermore, strategies on curriculum innovation will be discussed. Emphasis on curriculum process should be laid on the various curriculum, models i.e. whiller's models, Talars models, etc

EDU 211 Subject Methods

This course focuses on methods on teaching with specific tropics in student area of specialization. Various teaching methods will be discussed i.e. plenary methods, Socratic methods, lecture methods, team teaching methods, etc should be thought. Furthermore, the act of designing a lesson plans and lesson notes should also be taught. Students should also be expose to micro teaching to test their proficiency in the act of using instructional techniques available to them.

EDU 212 Measurement and Education

Basic principles used in measurement and evaluation should be thought. Statistical method used in analysis of data should also be taught. Different types of data and methods of analyzing each should also be taught.

EDU 213 Sociology of Education

Sociology principles involved in education are to be discussed. Emphasis should be laid on culture, society, factors binding society together, social stratification, social mobility, and factors

affecting social mobility Agencies of education together with some abnormal or deviant behaviour in the school system will also be discussed.

EDU 300 Teaching Practice

Students will be sent out for six weeks teaching practice exercise to put into practice the knowledge they have gained on course like developmental Psychology, philosophy of education, educational technology as well as educational psychology.

EDU 301 Educational Administration and Planning

Basic principles and theories of administration will be discussed. Planning in terms of programme planning of supervision strategies and general administration of schools will be discussed with emphasis on the duties of a principal or headmaster, the school inspectors, ministry of education, the function of the class teacher, inspectorate division of ministry of education in relation to supervision of instruction should be discussed.

EDU 302 Educational Technology

Historical background of educational technology, relevance of the use of technology to the business of teaching and learning should be dismissed. Educational media, the use of media and the relevance of media in teaching and learning situation should be emphasized. Two and three-dimensional aids, soft and hardware, projected and non-projected aids should also be discussed. Factors militating against the use of projected aids in Nigeria should also be discussed. Furthermore, improvisation of instructional materials should be taught.

EDU 311 Educational Psychology

Various Psychological principles involves in teaching and learning should be examined e.g. Pavlov's experiment, Thorndike, Skinner, etc. the effect of these psychological theories on teaching and learning process should be x-rayed. Individual differences and psychological make-up of the individual in relation to teaching and learning should also be examined.

EDU 312 Research Methods and Data Processing

This course will examine the meaning if research, processing involve in carrying out research. E.g. concept like research design, research problem, formulation of research hypothesis or research questions, development and validation of research instrument, methods of collecting research data and data analysis should also be discussed. Students should be taught how to write simple research reports from either experimental or quasi-experimental studies carry out by them or from a descriptive survey. They should also be taught how to acknowledge other authorities site in their work.

VTE 400 Research Project

Students are expected to carry out independent studies in their own field of specialization. Such studies are expected to focus on finding solutions to the problems of teaching and learning.

EDU 401 Practical Teaching

Students are expected to spend six weeks on industrial attachment to afford them of the opportunity on the job training in their area of specialization. Such training will expose them to modern techniques of administtry, or have an insight into what to expect in the world of work.

Such attachment also offers the student the opportunity to put into practice the knowledge, principle and practice they earlier acquired in the classroom during such attachment.

EDU 402 Guidance and Counseling

Definition of Guidance and counseling service, counseling processes, qualities of a good counselor occupation guidance, married counseling, counseling services in schools, contemporary issues in counseling services in Nigeria.

EDU 403 Continuous Assessment

Meaning of continuous assessment, national policy on education, its emphasis on continuous assessment, the domain of continuous assessment i.e. cognitive, Psychomotor and effective domains. The use or mis-use of continuous assessment in primary and post primary schools in Nigeria. Problems and prospects of the use of continuous assessment in schools in Nigeria.

EDU 411 Organization Of Primary And Secondary Schools In Nigeria

Programme organization and policy organization of primary schools in Nigeria, emphasis on the role of ministry of education, post-primary Education Board, Local Education Authority in Policy formulation and implementation in primary and secondary education. Contemporary issues and problems plaguing primary and secondary education in Nigeria.

EDU 413 Comparative Education

This course is designed to acquaint students with the various policies and programme of different countries, towards this end emphasis will be placed on British educational system, America, Ghana, Cote d'ivoir, China, Tanzania, Nigeria e. t. c.